

Towards a Mixed Reality Agent to Support Multi-Modal Interactive Mini-Lessons That Help Users Learn Educational Concepts in Context

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ABSTRACT

The discipline of education and training is in a state of transformation currently and is shifting toward more online learning environments as such, researchers are beginning to explore and apply new emerging technologies for online education. However, the current online learning platforms face new challenges in presenting learning content. There is a need for ways to support self-directed learning using new technological resources to create personally meaningful curiosity-driven learning experiences. In this work, an architectural framework for a mixed reality learning system is designed and presented to address this need. This is presented as an approach toward building tools for future educators to enable them to create multi-modal interactive mini-lessons in mixed reality that students can experience when they are within the appropriate learning contexts. Two proof-of-concept use case scenarios are presented using this framework. Together, this provides a step toward future multi-modal interactive educational agents that help users learn educational concepts in mixed reality.

1 MOTIVATION

Emerging technologies for education, such as AI (Artificial Intelligence), MR (Mixed Reality), and IoT (Internet-of-things) are advancing rapidly [1] and possess the potential to fulfill the requirement of fostering curiosity among learners [2] and supporting self-directed learning [4]. As these technologies converge they will enable new forms of context-aware agent system monitors using cloud-based infrastructure, as well as new forms of visualization [3]. Combining sources such as ego-centric sensing, environment cameras, and other wearable sensors makes it possible to gather context and visualize or augment educational content through MR head mounted displays. Together, this may enable learning experiences where users can build and discover contextual knowledge about their environment and drive personalized lessons based on their interests. To enable these experiences, context-driven and metaverse educational agent systems need to be designed to support users in learning, awareness, and knowledge management.

2 DESIGNING MIXED REALITY MULTI-MODAL MINI-LESSONS

2.1 Architecture

We present an architectural framework [Figure 1] to support a new form of learning and teaching supported by MR multimodal mini-lessons. For educators, this architecture provides support by allowing them to create new mini-lessons based on 3D user interface templates for a variety of educational topics. The architecture also supports self-driven learning in context through the use of 3D visualizations, access to the educational content, and conversational

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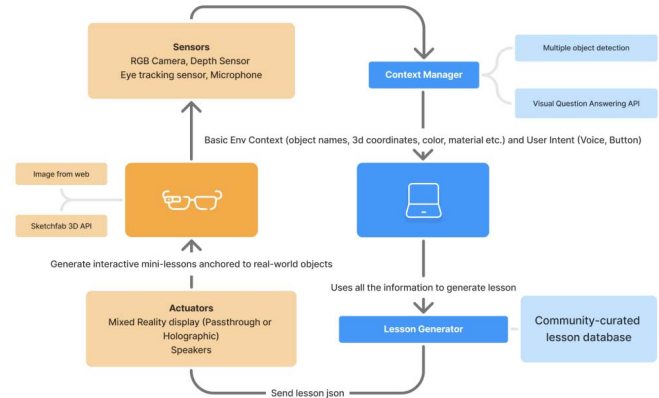


Figure 1: An architecture for MR multimodal interactive mini-lessons which combines head mounted displays, external sensors, conversational agents, 3D user interfaces, and machine learning models.

agents which allow them to explore further into the topic by querying the system. This is achieved by a combination of head mounted displays, external sensors, conversational agents, 3D user interfaces, and machine learning models. This architectural framework has been instantiated for the proof-of-concept use cases below involving mathematics and biology.

2.2 Mini-lesson 1: Biology

The first scenario presents an example case for a biology mini-lesson [Figure 2] experience. In this experience, the user learns about plant cells and interacts with the multi-modal MR framework to self-direct the learning content. This experience is initiated when the user encounters a plant in their environment.

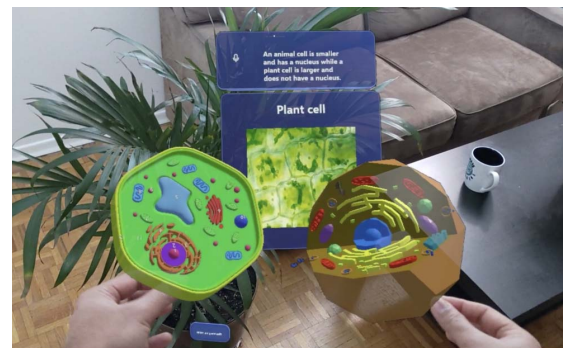


Figure 2: An example use-case based on the architecture for biology education showing the Plant Cell topic.

2.3 Mini-lesson 2: Mathematics

In the second scenario, an example case for learning a mathematics concept [Figure 3] is presented. This experience initiates a lesson about the volume of a cylinder when the user sees a mug. They are able to move, scale, and rotate the cylinder model to understand the underlying mathematical concepts in relation to the mug. They can also further engage with the multi-modal MR framework for a self-directed learning experience.

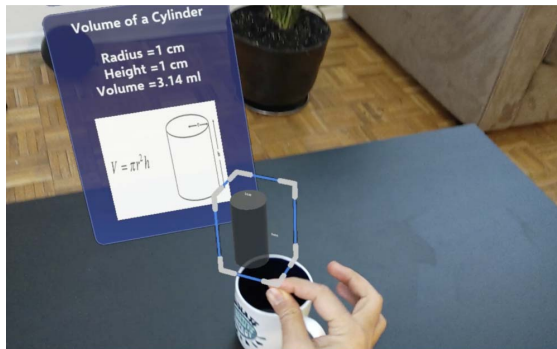


Figure 3: An example use-case based on the architecture for mathematics education showing the Volume of a Cylinder topic.

3 SUMMARY

This work has presented a multi-modal MR framework for educational concepts and two proof-of-concept use case scenarios. These use cases highlight how the underlying system can be used to create context-driven learning experiences in MR. These techniques and designs are envisioned to support the creation of many such lessons for different educational concepts across Science, Mathematics, and Languages. It is hoped that this research will inspire others in the XR community and advance the use of MR and AI hybrid designs for the future of curiosity-driven education.

REFERENCES

- [1] V. Chang and D. Fisher. The validation and application of a new learning environment instrument for online learning in higher education. In *Technology-rich learning environments: A future perspective*, pp. 1–20. World Scientific, 2003.
- [2] J. M. Glenn. Teaching the net generation. In *Business Education Forum*, vol. 54. ERIC, 2000.
- [3] A. Morris, J. Guan, N. Lessio, and Y. Shao. Toward mixed reality hybrid objects with iot avatar agents. In *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 766–773. IEEE, 2020.
- [4] G. Rozinaj, M. Vančo, R. Vargic, I. Minárik, and A. Polakovič. Augmented/virtual reality as a tool of self-directed learning. In *2018 25th International Conference on Systems, Signals and Image Processing (IWSSIP)*, pp. 1–5. IEEE, 2018.